

Haoran (Howard) Ding

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EDUCATION

University of California, Los Angeles

B.S. Electrical Engineering, Samueli School of Engineering | GPA: 3.74/4.00

Los Angeles, CA

Expected June 2029

Relevant Coursework: CS 31 & CS 32 (Programming in C++ I & II), Math 33A (Linear Algebra), Math 33B (Differential Equations), CS M51A (Logic Design of Digital Systems), ECE 102 (Systems and Signals)

RESEARCH EXPERIENCE

UCLA Summer Undergraduate Research Program (SURP)

Research Fellow — Terahertz and Infrared Photonics Group

Los Angeles, CA

June 2026 – Present

- Developed a MATLAB test automation suite communicating with lab instrumentation over GPIB to control and sequence a multi-instrument setup (step-scan FTIR spectrometer, function generator, lock-in amplifier) that replaced the manual test configuration via LabVIEW.
- Wrote scripts to automate instrument communication and full device characterization sequences — biasing the device under test and capturing output across pulsed and continuous-wave operating conditions — eliminating manual reconfiguration and improving test repeatability.
- Wrote data analysis scripts to process acquired test data, generate performance plots, and extract key device parameters (threshold current density, slope efficiency, characteristic temperature) against expected operating behavior.
- Built a GUI enabling other lab members to configure test parameters and run standardized validation sequences without editing underlying code, maintained under Git version control to support clear documentation and repeatable test procedures across the team.
- Independently brought up and characterized a previously untested hardware device — a metal-metal waveguide terahertz quantum cascade laser — establishing its full operating envelope: 2.45–2.61 THz bias tuning range and 133 K maximum operating temperature.

University of California, Santa Barbara

Student Researcher

Santa Barbara, CA

June 2024 – July 2024

- Engineered bio-inspired snake-scale prototypes to minimize tribological friction, iteratively redesigning interlocking mechanisms to resolve critical 3D-printing tolerance failures.
- Quantified performance using MATLAB and motion-tracking software to extract displacement and velocity datasets, accelerating design iteration cycles.

EXPERIENCE

Handshake AI

AI Training Fellow

Remote

January 2026 – Present

- Evaluate and refine frontier Large Language Models via structured Reinforcement Learning from Human Feedback (RLHF) pipelines, improving model reasoning quality and output accuracy.
- Audit complex AI-generated technical solutions for logical consistency across engineering and CS domains, providing calibrated feedback that informs model fine-tuning.

Bruin Underwater Robotics (ASME) at UCLA

Robotics Engineer

Los Angeles, CA

September 2025 – Present

- Contributed to AUV chassis frame architecture and thruster configuration as part of the mechanical design team, evaluating trade-offs in rigidity, buoyancy distribution, and hydrodynamic stability while collaborating with the electrical subsystem team on component integration.

MATE ROV Underwater Robotics

Team CEO

San Jose, CA

September 2022 – May 2025

- Led a multidisciplinary team through the full design-build-pilot lifecycle of a Remotely Operated Vehicle, earning 1st Place Overall and 1st Place in Engineering & Communication at the 2023 MATE ROV Monterey Regional.
- Spearheaded CAD development of critical ROV components in Onshape, applying hydrodynamic design principles and iterative 3D-printing prototyping to minimize structural weight and improve efficiency.

PROJECTS

AI-Powered Market Analyst Agent

Python | Gemini API | Financial Modeling Prep API

2026

- Architected a multi-source financial intelligence pipeline in Python integrating the Gemini and Financial Modeling Prep APIs, with edge-case logic that correctly excludes banking institutions from DCF valuation.
- Automated end-to-end investment memo generation — from raw financial data to structured valuation and risk analysis — cutting research time from hours to seconds.

Quantum RAG Assistant

Python | ChromaDB | LLM Integration | React/TypeScript | Langgraph

2026

- Built a Retrieval-Augmented Generation system with ChromaDB to ingest, parse, and query semiconductor physics literature, accelerating literature review for ongoing lab research; implemented as a LangGraph agent with a self-critique retry loop returning cited sources.
- Converted the tool into a full-stack web app with a React/TypeScript frontend deployed via CI/CD, and production-style backend safeguards (auth, rate limiting, CORS) with persistent multi-turn conversation state.

Autonomous Line-Following Robot

C++ | Arduino | PID Control

2026

- Designed and tuned a closed-loop PID controller on an Arduino, integrating an IR sensor array with motor driver circuitry; empirically tuned gain coefficients to minimize tracking error and oscillation across variable-contrast surfaces.

TECHNICAL SKILLS

Languages: MATLAB, Python, C++, C, Java, git, Typescript,

Instrumentation: Instrument Control & Automation, FTIR Spectroscopy, Lock-In Amplifiers, Function & Pulse Generators, Cryostats, LabVIEW, MATLAB GUI Development, GPIB communication, oscilloscopes

AI & Software: Gemini API, RAG, ChromaDB, RLHF, React, REST API Design, CI/CD, LangGraph, Vitest, React Testing Library

Hardware & Design: Arduino, STM32, Circuit Analysis, Soldering, CAD, 3D Printing